|elvtr|

ASSIGNMENT #5: VEGETATION INDEX (NDVI) CALCULATION AND CROP HEALTH ANALYSIS

ASSIGNMENT OVERVIEW

FIELD	VALUE
ASSIGNMENT NUMBER	5
TITLE	Vegetation Index (NDVI) Calculation and Crop Health Analysis
GRADE VALUE	8 points
DUE DATE	March 16, 2026, 11:59 PM PT

Objectives

After completing this assignment, you will be able to:

- Use skills to run one satellite-imagery read workflow
- Generate one NDVI output image
- Capture cloud-masking/combination context at a basic level
- Save reproducible input + script + output assets

Overview

This assignment is intentionally narrower because Class 7 is harder. Focus on one solid end-to-end example: load imagery, compute NDVI, generate one clear NDVI image, and preserve reproducibility artifacts.

Instructions

Step 1: Start from the Class Template and Import Skills

Use the same setup links as Assignments 1-3:

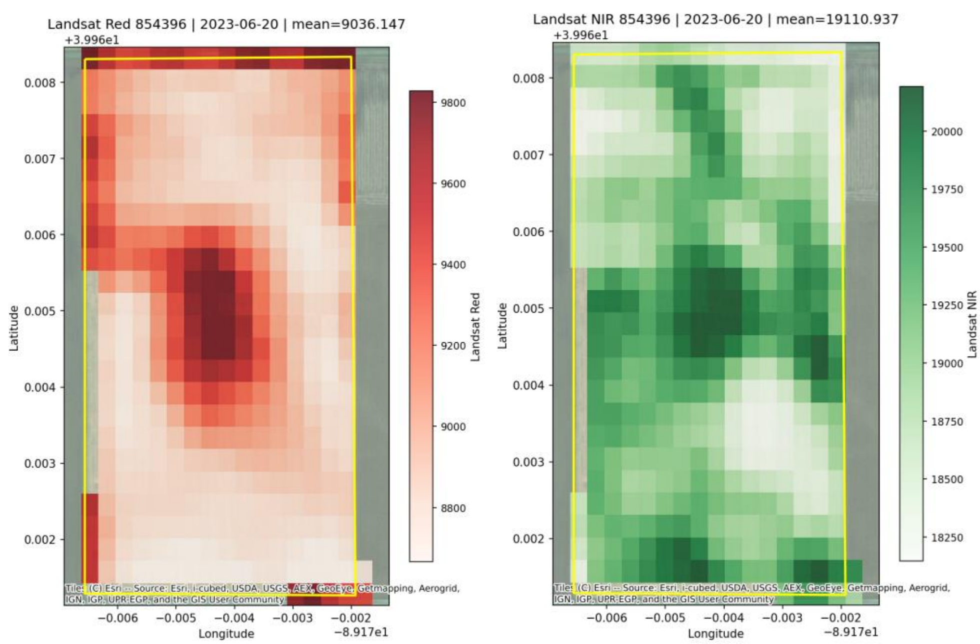
- Template branch: https://github.com/SuperiorByteWorks-LLC/agent-project/tree/feat/agri-skills-for-agent-project
- Skills source: https://github.com/borealBytes/ag-skills/tree/skills-content

In OpenCode, paste:

```
Import skills from https://github.com/borealBytes/ag-
skills/tree/skills-content into this current project and make them
available for OpenCode to use.
```

Mark complete: ☐

Step 2: Run One Data-Read Example with Skills

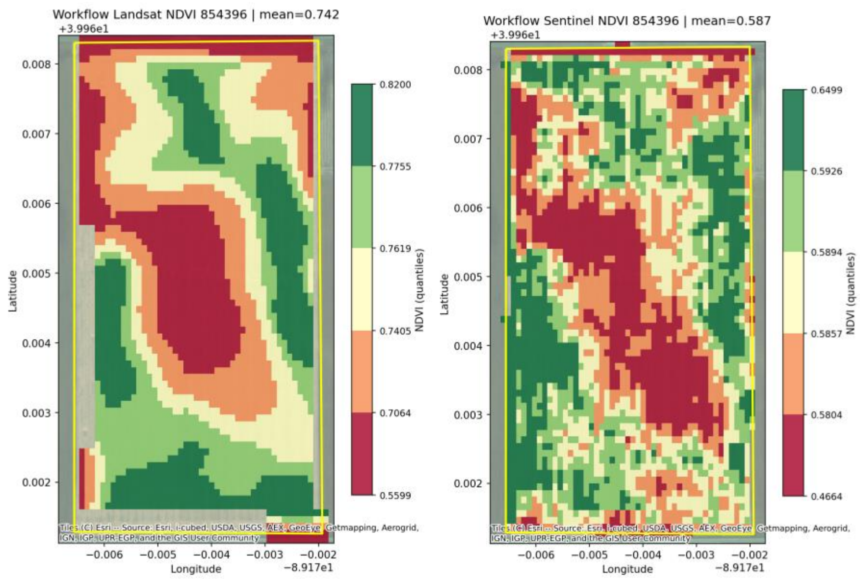


Use Class 7 skills to read/load one imagery example and save your input artifacts under data/assignment-05/.

At minimum, produce one **single-band output** image (for example, red or NIR) so we can clearly see your data-read step.

Mark complete: ☐

Step 3: Generate One NDVI Image



Produce one NDVI output image from your loaded data.

If you successfully produce NDVI, that also shows your band-combination step worked.

Your map should include:

- clear title
- readable legend
- clean visual presentation

Mark complete: ☐

Step 4: Add a Short Markdown Walkthrough with Inline Images

Create a short markdown file in your repo (for example reports/assignment-05-walkthrough.md) that includes:

- what you ran,
- one single-band image inline,
- one NDVI image inline,

- a short note on what each image shows.

Keep this concise and classroom-readable.

Mark complete: ☐

Step 5: Document Basic Cloud/Combination Context

Add a short note in your repo indicating what cloud-masking or sensor-combination step was used (or intentionally deferred).

Keep this lightweight; the core requirement is one successful NDVI output.

Mark complete: ☐

Step 6: Capture Inputs + Scripts + Outputs

Keep all reproducibility artifacts in your data folder:

- inputs used
- script(s) used
- single-band output image(s)
- NDVI output image(s)

Mark complete: ☐

Progress Checklist

- ☐ Class template + skills import setup completed
- ☐ One imagery data-read run completed
- ☐ One single-band output image generated
- ☐ One NDVI output image generated
- ☐ Markdown walkthrough with inline images added
- ☐ Cloud/combination context note added

- ☐ Inputs/scripts/outputs captured

Deliverables

Required

Deliverable	Description
Repository/branch link	Link to branch containing Assignment 5 run assets
Run evidence	1-2 screenshots showing successful execution + NDVI output image
Single-band artifact	At least one single-band output image in repo
NDVI image artifact	At least one NDVI output image in repo
Markdown walkthrough	Short markdown walkthrough with inline single-band + NDVI images

Submission Format

- Submit via: Google Classroom or course-designated submission portal
- File formats: screenshots (.png/.jpg) + branch/repo link
- Optional: If you include extra notes/evidence in your repo, add one short note in Google Classroom telling the instructor where to look.

Grading Criteria

Criterion	Points	Description
Repository evidence	5	Branch/repo includes Class 7 assignment assets and

Criterion	Points	Description
		reproducibility files
NDVI output quality	3	NDVI image is present, readable, and clearly generated from workflow
Run evidence quality	2	Screenshots clearly prove successful execution

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Key Reminders

- Keep this assignment focused and reproducible.
- One working NDVI example is enough for full scope.
- Use skills for execution and keep artifacts organized.
- Classroom focus: simple, clear proof that you can read imagery and generate NDVI.
